Property	Boron	Aluminum	Gallium	Indium	Thallium
Atomic Mass ($g.mol^{-1}$ at 293 K)	10.81	26.98	69.72	114.81	204.38
Electronic configuration	$[He]2s^2\ 2p^1$	$[Ne]3s^2\ 3p^1$	$[Ar]3d^{10}\ 4s^2\ 4p^1$	$[Kr]4d^{10}\ 5s^2\ 5p^1$	$[Kr]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^1$
Atomic radius (Å)	1.92	1.84	1.87	1.93	1.96
Density ($g.cm^{-3}$ at 293 K)	2.34	2.70	5.91	7.31	11.80
Melting point (K)	2350	933	302.76	429	577
Boiling point (K)	4273	2792	2502	2300	1746

2.2.3 Chemical properties of boron:

Boron is the only nonmetal in this group and is less reactive. However, it shows reactivity at higher temperatures. Many of its compounds are electron deficient and has unusual type of covalent bonding which is due to its small size, high ionisation energy and similarity in electronegativity with carbon and hydrogen.

Formation of metal borides:

Many metals except alkali metals form borides with a general formula M_xB_y (x ranging upto 11 and y ranging upto 66 or higher)

Direct combination of metals with boron:

$$Cr + nB \xrightarrow{1500\ K} CrB_n$$

Reduction of borontrihalides:

Reduction of borontrichloride with a metal assisted by dihydrogen gives metal borides.

$$BCl_3 + W \xrightarrow[H_2]{1500\ K} WB + Cl_2 + HCl$$

Formation of hydrides:

Boron does not react directly with hydrogen. However, it forms a variety of hydrides called boranes. The simplest borane is diborane - B_2H_6. Other larger boranes can be prepared from diborane. Treatment of gaseous boron trifluoride with sodium hydride around 450 K gives diborane. To prevent subsequent pyrolysis, the product diborane is trapped immediately.

$$2BF_3 + 6NaH \xrightarrow{450\ K} B_2H_6 + 6NaF$$

Formation of boron trihalides:

Boron combines with halogen to form boron trihalides at high temperatures.

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